

Supplementary Information

Transition-state damping as a mechanistic descriptor of rate turnover in chemical reactivity

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S0.1 The exceptional point at $\chi = 1$

The second-order generator $\ddot{q} + \Gamma \dot{q} + \Omega^2 q = 0$ is equivalent to the first-order system $d\mathbf{x}/dt = A\mathbf{x}$ with state vector $\mathbf{x} = (q, \dot{q})^\top$ and matrix

$$A = \begin{pmatrix} 0 & 1 \\ -\Omega^2 & -\Gamma \end{pmatrix}. \quad (\text{S1})$$

The eigenvalues are $\lambda_{\pm} = -\Gamma/2 \pm \sqrt{\Gamma^2/4 - \Omega^2}$. For $\chi \neq 1$ these are distinct and A is diagonalizable over

C. At $\chi = 1$, eigenvalues coalesce at $\lambda_+ = \lambda_- = -|\Omega|$, the geometric multiplicity drops to 1, and A is defective with Jordan normal form

$$J = \begin{pmatrix} -|\Omega| & 1 \\ 0 & -|\Omega| \end{pmatrix}, \quad (\text{S2})$$

which is the defining signature of a second-order non-Hermitian exceptional point [17]. The time-domain Green's function changes class at this point:

$$\chi < 1: \quad h(t) = e^{-\Gamma t/2} \sin(\Omega_d t) / \Omega_d \quad (\text{oscillatory; recrossing}) \quad (\text{S3})$$

$$\chi = 1: \quad h(t) = t e^{-|\Omega| t} \quad (\text{polynomial-exponential; critical}) \quad (\text{S4})$$

$$\chi > 1: \quad h(t) = C_+ e^{\lambda_+ t} + C_- e^{\lambda_- t} \quad (\text{biexponential; overdamped}) \quad (\text{S5})$$

The long-time decay rate for $\chi > 1$ is controlled by the less-negative eigenvalue:

$$|\lambda_-| = |\Omega| \left(\chi - \sqrt{\chi^2 - 1} \right). \quad (\text{S6})$$

The function $f(\chi) = \chi - \sqrt{\chi^2 - 1}$ is strictly decreasing for $\chi > 1$ with $f(1) = 1$ and $f(\chi) \rightarrow 0$ as $\chi \rightarrow \infty$. Among all parameter choices producing purely monotone decay, $\chi = 1$ therefore maximizes the decay rate: fastest commitment without oscillation.

S0.2 The dynamical efficiency curve and the 5% window

Define dynamical efficiency as $\eta(\chi) = k(\chi)/k_{\max}$, where $k(\chi)$ is the Pollak–Grabert–Hänggi (PGH) interpolation rate and $k_{\max}$ is the maximum over all χ at fixed thermodynamic parameters. The PGH rate combines two limits:

- Energy-diffusion limit ($\chi \ll 1$): $k_{\text{ED}} \propto \chi$ (rate increases with friction).
- Spatial-diffusion limit ($\chi \gg 1$): $k_{\text{SD}} \propto 1/\chi$ (rate decreases with friction).

Their crossover produces a rate maximum that shifts with reduced barrier height $V^\ddagger/k_B T$. Over the physically relevant range $V^\ddagger/k_B T \in [5, 30]$, the maximum falls within $\chi \in [0.7, 1.3]$. The window $[0.8, 1.2]$ corresponds to $\eta \geq 0.95$ for $V^\ddagger/k_B T \geq 8$.

A note on the Kramers transmission coefficient: the Kramers–Smoluchowski coefficient in the high-friction limit is $\kappa_{\text{Kr}} = \sqrt{1 + \chi^2} - \chi$, which is strictly monotonically decreasing from 1 to 0 and has no maximum. The Kramers turnover arises from the combined PGH rate (energy-diffusion plus spatial-diffusion regimes), not from κ_{Kr} alone. κ_{Kr} should not be presented as having a maximum at $\chi = 1$.

S0.3 Connection to Kramers, Grote–Hynes, and Zusman

Grote–Hynes extension. In non-Markovian baths, the effective χ becomes frequency-dependent: $\chi_{\text{eff}}(\Omega) = \hat{\gamma}(\Omega)/(2\Omega_{\text{TS}})$, where $\hat{\gamma}(\Omega)$ is the Laplace-transformed memory kernel at the reactive frequency. For exponential memory $K(t) = \gamma_0 e^{-t/\tau_{\text{mem}}}$:

$$\hat{\gamma}(\Omega) = \frac{\gamma_0}{\sqrt{1 + (\Omega\tau_{\text{mem}})^2}}. \quad (\text{S7})$$

The $\chi = 1$ separatrix broadens into a band of width $\sim \Omega\tau_{\text{mem}}$ but remains well-defined for $\Omega\tau_{\text{mem}} < 0.5$.

Zusman theory. The adiabaticity parameter $g = 2\pi V^2\tau_L/(\hbar\sqrt{4\pi\lambda k_B T})$ scales as $g \sim 2\chi_{\text{ET}}$ to within a prefactor that evaluates to 0.67–1.0 at ambient temperature for $\lambda \in [0.5, 1.0]$ eV. These two quantities are distinct and must not be conflated:

Quantity	Physical meaning	Tunes
g (Zusman)	Electronic coupling adiabaticity	Role of H_{ab} (electronic coupling matrix element)
χ	Solvent friction damping	Role of τ_L (nuclear dynamics)

Table S1: Distinction between the Zusman adiabaticity parameter g and the CCE damping ratio χ . Both conditions must be satisfied simultaneously for efficient concerted electron transfer.

The mapping $g \sim 2\chi$ is specific to ambient-temperature ET with standard reorganization energies. At cryogenic temperatures or with very small or large λ , the full temperature-dependent expression is required.

S1 Detailed Bias Analysis by Estimator Route

S1.1 Route A: spectroscopic proxy

The connection between observed linewidth and the true dissipation rate is:

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_2^*} \implies \Delta\tilde{\nu}_{\text{obs}} = \underbrace{\Delta\tilde{\nu}_L}_{\text{homogeneous}} + \underbrace{\Delta\tilde{\nu}_G}_{\text{inhomogeneous}} \quad (\text{S8})$$

Route A yields $\chi_{\text{spec}} = \Delta\tilde{\nu}_L/(2\tilde{\nu})$, which equals the true χ_{TS} only when $T_2 \approx 2T_1$ (homogeneous lifetime broadening dominates).

The χ_{spec} values cited for Fe(111) and Fe(110) in N_2 activation derive from DFT-computed electronic friction linewidths Γ_{cat} (Maurer et al. 2017), not from raw operando IR linewidths; these are Route C inputs interpreted as Route A estimates. $\chi_{\text{spec}} \approx \chi_{\text{DFT}}$ only when homogeneous broadening dominates the measured spectrum, a condition that must be verified experimentally. Until operando T_1 measurements are available for these systems, the reported χ_{spec} values carry the Route C bias described in Section S1.3.

Table S2 summarizes sources of linewidth inflation and their mitigations.

Source	Magnitude	Diagnostic	Mitigation
Thermal Doppler broadening (surface)	20–50 cm ⁻¹ at 400°C	Gaussian profile component	Measure at low T ; extrapolate Lorentzian to $T \rightarrow 0$
Inhomogeneous site distribution	Varies with surface prep	Coverage-dependent linewidth	Single-crystal at low coverage; verify $\Delta\tilde{\nu}$ vs. θ
Pure dephasing (T_2^*)	Coupling-dependent	Voigt Gaussian fraction > 50%	Extract Lorentzian component only; cross-check with T_1
Instrument lineshape	Apodization artifact	Deviation from Lorentzian	Deconvolve instrument function; high-resolution FTIR

Table S2: Sources of linewidth inflation in Route A and their mitigations.

S1.2 Route B: m_{eff} uncertainty propagation

The propagation from m_{eff} uncertainty to χ uncertainty follows from:

$$\Omega = \sqrt{\frac{2\lambda}{m_{\text{eff}} (\Delta Q)^2}} \implies \chi \propto m_{\text{eff}}^{+1/2} \quad (\text{S9})$$

A decade uncertainty in m_{eff} (50 to 500 amu) shifts χ by a factor of $\sqrt{10} \approx 3.2$. The sensitivity analysis in Table S3 demonstrates robustness of regime assignment despite this absolute uncertainty.

m_{eff} (amu)	χ_{ET} (TEMPO/MeCN, $\tau_L = 0.2$ ps)	Regime
50	0.15	Underdamped
100	0.11	Underdamped
200	0.076	Underdamped
500	0.048	Underdamped

Table S3: Route B sensitivity to m_{eff} for TEMPO in acetonitrile. All values across the decade span fall in the underdamped regime; the regime assignment is robust.

All values across the decade span of m_{eff} fall well below 0.8. The regime assignment (underdamped, approaching-critical) is robust regardless of the specific m_{eff} assumed. The absolute value $\chi_{\text{ET}} \approx 0.11$ carries no quantitative significance in isolation.

For relative χ , m_{eff} cancels exactly when the same donor–acceptor pair is compared across solvents: $\chi_{\text{ET}}^{(A)}/\chi_{\text{ET}}^{(B)} = \tau_L^{(A)}/\tau_L^{(B)}$. For TEMPO, $\chi_{\text{PC}}/\chi_{\text{MeCN}} = 1.20/0.20 = 6\times$, mapping the full range from near-acetonitrile values to near-propylene carbonate values without any m_{eff} assumption.

S1.3 Route C: DFT friction bias

The Newns–Anderson expression $\Gamma_{\text{cat}} = 2\pi|V_k|^2\rho(\varepsilon_F)$ is a mean-field, single-particle result [18,19]. Three sources of systematic error require attention.

For 3d ferromagnets, GGA-DFT underestimates d-band exchange splitting in Fe, Co, and Ni. The predicted Γ_{cat} can be off by 20–50%. For quantitative predictions on these metals, DFT+ U with $U \approx 2\text{--}4$ eV for Fe is recommended; Route A provides a model-independent cross-check.

Route C computes χ at the adsorbate ground-state geometry and is therefore a lower bound on χ_{TS} : as π^* -backdonation increases and Ω_{bond} decreases as the bond stretches toward the transition state, χ_{DFT} understates χ_{TS} in most catalytic systems. For quantitative transition-state predictions, γ_{NN} and the harmonic frequency should be recomputed at the saddle-point geometry.

Spin-orbit coupling is negligible for light adsorbates (N, O, CO) on 3d metals but becomes relevant for heavy adsorbates on Pd, Pt, and Au; non-collinear DFT should be included if the adsorbate is heavy.

Cross-check: the DFT-predicted IR linewidth should be $\Delta\tilde{\nu} = \hbar\Gamma_{\text{cat}}/hc$ in cm^{-1} . Agreement with the Lorentzian linewidth from Route A within a factor of two confirms self-consistency.

S2 Domain-Specific Derivations

S2.1 Electron transfer: mapping to Marcus–Zusman

The Marcus solvent reorganization coordinate Q obeys:

$$m\ddot{Q} + \gamma_{\text{solv}}\dot{Q} + k(Q - Q_{\text{min}}) = \xi(t), \quad k = \frac{2\lambda}{(\Delta Q)^2} \quad (\text{S10})$$

Identifying terms with the minimal generator (note this is a stable oscillator describing solvent coordinate motion in the precursor well; the barrier-top curvature enters through Ω_{TS} in the rate expression): $\Omega^2 = k/m = 2\lambda/(m_{\text{eff}}\Delta Q^2)$; $\Gamma = \gamma_{\text{solv}}/m \approx 1/\tau_L$. Therefore:

$$\chi_{\text{ET}} = \frac{1/\tau_L}{2\sqrt{2\lambda/(m_{\text{eff}}\Delta Q^2)}} \quad (\text{S11})$$

The Zusman adiabaticity parameter $g = 2\pi V^2\tau_L/(\hbar\sqrt{4\pi\lambda k_B T})$. Substituting $\tau_L = 1/\Gamma$ and using Ω_{solv} :

$$\frac{g}{2\chi_{\text{ET}}} = \frac{\pi V^2}{\hbar\Omega_{\text{solv}}\sqrt{\pi\lambda k_B T}} \quad (\text{S12})$$

At $T = 300$ K, $\lambda = 0.7$ eV, $V = 200$ cm^{-1} , this prefactor evaluates to approximately 0.8, confirming $g \sim 2\chi$ to within order unity at standard conditions. The mapping $g \sim 2\chi$ does not mean g and χ are the same quantity; see Section S0.3 for the distinction.

Why the Kramers turnover maximum structurally implies $\chi \approx 1$

The PGH combined rate operates in two opposing limits: the energy-diffusion branch gives $k_{\text{ED}} \propto \Gamma$ (rate rises with friction because the bath supplies activation energy) and the spatial-diffusion branch gives $k_{\text{SD}} \propto 1/\Gamma$ (rate falls with friction because viscous drag slows barrier transit). Their crossover defines the rate maximum. In the spatial-diffusion limit, the Kramers transmission coefficient for escape over a parabolic barrier is

$$\kappa_{\text{Kr}} = \sqrt{\left(\frac{\Gamma}{2\omega_b}\right)^2 + 1} - \frac{\Gamma}{2\omega_b}, \quad (\text{S13})$$

which has its steepest descent (the knee marking the onset of strong friction suppression) at $\Gamma \approx 2\omega_b$, i.e., $\chi = \Gamma/(2\omega_b) \approx 1$. The full PGH rate maximum occurs near this knee, shifted slightly by the energy-diffusion branch weight and the reduced barrier height $V^\ddagger/k_B T$. Consequently $\chi \approx 1$ at the turnover is not a numerical coincidence: it is a structural consequence of the two-regime form of barrier-crossing theory. Any system whose kinetics exhibit a Kramers turnover will have its rate maximum near $\chi = 1$ by the same argument, regardless of the specific chemistry. This is the structural reason the TEMPO solvent series, the Route B ET closure ($\chi_{\text{ET}} = 0.992$, Section S4), and the catalytic promoter-loading curves all locate their dynamical optima within the same dimensionless window.

S2.2 Catalysis: Newns–Anderson to χ

An adsorbate on a metal surface has its vibrational level ε_a broadened by hybridization with metal continuum states:

$$\Gamma_{\text{cat}} = 2\pi|V_k|^2\rho(\varepsilon_F) \quad [\text{eV}] \quad (\text{S14})$$

Converting to a damping ratio comparable to the bond frequency:

$$\chi_{\text{DFT}} = \frac{\Gamma_{\text{int}} + \Gamma_{\text{cat}}}{2hc\tilde{\nu}_{\text{bond}}} \quad hc = 1.23984 \times 10^{-4} \text{ eV}\cdot\text{cm} \quad (\text{S15})$$

The K-promoter effect on Fe(111) operates simultaneously on both levers. K donates electron density, increasing $\rho(\varepsilon_F)$ and hence Γ_{cat} (raising Γ). Additional d-band backdonation into $\text{N}_2 \pi^*$ reduces $\tilde{\nu}$ from 2359 cm^{-1} (gas phase) to 1415 cm^{-1} , decreasing Ω_{TS} (lowering Ω). Both effects increase χ toward unity, structurally accounting for the empirically observed activity enhancement.

S2.3 Fe(111) transition-state χ : analytical bracket

The Route C estimator applied to the ground-state geometry of K-promoted Fe(111) gives $\chi_{\text{DFT}} \approx 1.28$ using $\gamma_{\text{NN}} = 0.45 \text{ eV}$ and $\tilde{\nu}_{\text{NN}} = 1415 \text{ cm}^{-1}$ (adsorbed-state frequency confirmed by HREELS). This section provides the analytical bracket showing that χ_{TS} is well into the overdamped regime at the saddle point, consistent with the approach-dynamics interpretation in the main text.

Two levers that shift χ between ground state and transition state

Two changes occur simultaneously on the path from ground state to transition state, and both push χ upward.

Bond softening (Ω_{TS} decreases). The N–N bond stretches from approximately $1.18 \text{ \AA}$ (adsorbed precursor, γ -state on Fe(111)) toward the transition-state geometry. The 6D-PES calculations of Nosir et al. place the most reactive dissociation channels at TS bond lengths of $1.5\text{--}1.8 \text{ \AA}$. Using Morse-potential scaling $\nu \propto \exp[-a(r - r_e)]$ calibrated to the known frequencies at r_e (gas phase: 2359 cm^{-1} at $1.10 \text{ \AA}$; γ -state: 1415 cm^{-1} at $1.18 \text{ \AA}$), the N–N stretch frequency at the TS geometry extrapolates to:

$$\tilde{\nu}_{\text{TS}} \approx 400\text{--}900 \text{ cm}^{-1} \quad (r_{\text{TS}} = 1.5\text{--}1.8 \text{ \AA}) \quad (\text{S16})$$

Enhanced backdonation (γ_{NN} increases or holds). As the N–N bond elongates, π^* -backdonation from the Fe d-band increases, further broadening the N_2 vibrational level. The electronic friction at the TS is therefore $\geq \gamma_{\text{NN}}^{\text{GS}} = 0.45 \text{ eV}$. Taking the conservative lower bound $\gamma_{\text{NN}}^{\text{TS}} = \gamma_{\text{NN}}^{\text{GS}}$:

$$\chi_{\text{TS}} \geq \frac{0.45 \text{ eV}}{2 \times 1.24 \times 10^{-4} \text{ eV} \cdot \text{cm} \times 900 \text{ cm}^{-1}} \approx 2.0 \quad (\text{S17})$$

and with enhanced backdonation and the softer TS frequency:

$$\chi_{\text{TS}} \lesssim \frac{0.70 \text{ eV}}{2 \times 1.24 \times 10^{-4} \times 400 \text{ cm}^{-1}} \approx 7.1 \quad (\text{S18})$$

Physical reconciliation

The ground-state Route C gives $\chi_{\text{DFT}} \approx 1.28$, describing the approach dynamics in the precursor well. The TS bracket places $\chi_{\text{TS}} \in [2, 7]$ at the saddle point. This is not a contradiction: the framework does not require $\chi_{\text{TS}} \approx 1$ for this system. The physically meaningful claim is that approach-region χ governs whether the coordinate commits to the barrier or recrosses. For unpromoted Fe(111), $\chi_{\text{DFT}} \approx 0.10$ (deeply underdamped), so N_2 oscillates in the precursor well without committing and the transition state is rarely reached. For K-promoted Fe(111), $\chi_{\text{DFT}} \approx 1.28$ (near-critical approach), so N_2 approaches the barrier monotonically with minimal recrossing; once at the transition state, $\chi_{\text{TS}} \gg 1$ ensures fast monotonic descent to products. The PGH unified rate formula identifies the approach damping ratio, not the post-TS value, as what governs the rate turnover.

Direct experimental test

If the TS frequency is measured directly via reaction-path normal-mode analysis from the Nosir PES or via ML dynamics, the calculation above becomes exact. The predicted value is $\tilde{\nu}_{\text{TS}} \approx 400\text{--}900 \text{ cm}^{-1}$, corresponding to $\chi_{\text{TS}} \approx 2\text{--}7$. A measurement in this range confirms the framework; a measurement returning $\tilde{\nu}_{\text{TS}} > 1200 \text{ cm}^{-1}$ (requiring near-zero bond softening) would be inconsistent with the known TS geometry and would require revision.

S2.4 PCET: mechanism selection via χ_{eff}

In PCET, the coupled proton–electron coordinate has an effective dissipation including both solvent dielectric friction on the electron and coupling of the proton motion to bath phonons:

$$\chi_{\text{eff}} = \frac{\Gamma_{\text{ET}} + \Gamma_{\text{PT}}}{2\Omega_{\text{coupled}}} \quad (\text{S19})$$

H $\rightarrow$ D substitution increases m_{PT} by a factor of two and shifts Ω_{PT} by $\sqrt{2}$:

$$\chi_{\text{eff}}^{\text{D}} = \frac{\chi_{\text{eff}}^{\text{H}}}{\sqrt{2}} \approx 0.71 \chi_{\text{eff}}^{\text{H}} \quad (\text{S20})$$

If $\chi_{\text{eff}}^H \approx 1$ (concerted), then $\chi_{\text{eff}}^D \approx 0.71$ (approaching stepwise from below). The experimental signature is a KIE that rises then falls with increasing viscosity, reaching its maximum at the $\chi_{\text{eff}}^H = 1$ crossing (Test 4, Section S3). This KIE maximum is a dynamical signature of the $\chi = 1$ separatrix, not a tunneling signature, and is predicted in a fully classical framework.

S3 Falsification Protocol

The CCE framework is falsifiable. Each test in Table S4 defines experimental conditions, minimum dataset requirements, predicted outcomes, and a clear falsification criterion. A single well-controlled failure within the stated domain and scope falsifies the corresponding claim.

#	Domain	Measurement	Prediction	Falsified if	Min. dataset
1	ET	k_{et} vs. viscosity at fixed λ , ΔG°	Rate maximum at viscosity where $\chi_{\text{ET}} \approx 1$	Rate monotonic in viscosity; no maximum	≥ 5 viscosities; λ constant $\pm 5\%$
2	ET	Femtosecond TA vs. solvent across $\chi = 0.1$ –2	$\chi < 1$: oscillatory wavepacket; $\chi \approx 1$: spectral merging; $\chi > 1$: monotonic	No qualitative spectral transition near $\chi = 1$	≥ 3 solvents spanning full range
3	PCET	KIE, pH, intermediates vs. pressure 0–10 kbar	Stepwise $\rightarrow$ concerted switch expected when χ_{eff} approaches the critical window $[0.8, 1.2]$	Mechanism switch uncorrelated with χ	≥ 5 pressures; χ computed at each P
4	PCET	k_H/k_D vs. solvent friction at fixed T	KIE rises then falls; maximum at $\chi_{\text{eff}}^H \approx 1$; D shifted by $1/\sqrt{2}$	KIE friction-independent or monotonic	≥ 4 solvents; H and D measured separately
5	Catalysis	Volcano-plot outlier TOF vs. χ_{spec} OR χ_{DFT} across ≥ 5 metals/facets	Positive outliers at $\chi \approx 1$; negative outliers at $\chi \ll 1$ or $\gg 1$	Outlier activity correlates with binding energy alone	≥ 5 metals or facets with χ and TOF
6	Catalysis	$^{14}\text{N}/^{15}\text{N}$ dissociation ratio on Fe(111) vs. Fe(110)	Fe(111) shows larger isotope effect (mass shift moves χ further from window)	Isotope effects surface-independent	Both facets; both isotopes; same conditions
7	Catalysis	Operando $\Delta\tilde{\nu}_L/\tilde{\nu}$ and TOF vs. K-promoter loading	TOF maximizes at finite loading where $\chi_{\text{spec}} \approx 1$; over-promotion reduces TOF	TOF increases monotonically with $\Delta\tilde{\nu}_L/\tilde{\nu}$	≥ 3 promoter loadings; TOF $\pm 10\%$; Voigt-fitted linewidths
8	Cross-domain	χ at optimal conditions across ET, PCET, catalysis for independent systems	All optimal systems cluster within $\chi \in [0.8, 1.2]$; $p < 0.05$ vs. uniform prior on $\log_{10} \chi \in [-4, 1]$	Optimal χ scatter outside $[0.8, 1.2]$; no clustering	≥ 1 independent optimal system per domain

Table S4: Eight falsifiable predictions of the CCE framework. All tests must be performed within scope boundaries (thermally activated, near-Markovian, $\text{KIE} < 7$). Test 7 is the most experimentally direct original prediction.

Test 8: independence requirement and statistical note

Test 8 computes $p = (w/r)^N$ where $w = \log_{10}(1.2/0.8)$ and $r = \log_{10}(10^1/10^{-4}) = 5$ (log-uniform prior on accessible χ -space). This interval spans the physically accessible range from extremely underdamped molecular dynamics ($\chi \approx 10^{-4}$, e.g. gas-phase N_2) to strongly overdamped condensed-phase kinetics ($\chi \approx 10^1$, e.g. heavy-metal surface adsorbates), and assumes no prior knowledge of where optima should fall. For $N = 3$ independent systems: $p \approx (0.0352)^3 \approx 4.4 \times 10^{-5}$. The log-uniform prior choice does not affect the qualitative clustering conclusion; alternative prior choices shift the p -value by $\mathcal{O}(1)$ factors only, and the result is significant under any reasonable prior.

The three currently qualifying independent systems are: (1) the ET domain, TEMPO radical at its optimal-turnover solvent, where k_{et} is maximized in the rate-versus- $1/\tau_L$ series; (2) the PCET domain, a Ru-tyrosine or analogous PCET complex at the pressure or viscosity of maximum KIE, where mechanism switches from stepwise to concerted; (3) the catalysis domain, Fe(111) at optimal K-promoter loading where TOF is maximized (Test 7 operating point). For the ET domain, the damping ratio at the optimal turnover condition is computed explicitly from tabulated solvent parameters using $\chi_{\text{ET}} = 1/(2\tau_L\Omega) = 0.992$ (Section S4). For the PCET domain, the optimal operating condition is identified through the mechanistic activation-volume contrast (Langford et al. 2025 [ref28 in main text]): near-zero $\Delta V^\ddagger$ in the concerted regime confirms friction bypass at the near-critical damping condition, providing an independent empirical anchor for the $\chi_{\text{eff}} \approx 1$ boundary without requiring a direct χ_{eff} calculation from parameters not yet tabulated in published form. For the catalysis domain, $\chi_{\text{DFT}} \approx 1.28$ is computed from first-principles electronic friction tensors via Route C. The three systems therefore represent distinct evidence types: computed from solvent parameters, empirically anchored via activation volume, and computed from DFT friction, from three independent chemical families, independent both chemically and mechanistically: solvent-controlled ET, coupled PCET dynamics, and metal-surface catalysis. Systems that do not qualify: multiple solvents from the same donor-acceptor pair, multiple promoter loadings on the same surface, or chemically analogous systems from the same family. The statistical claim is meaningful only with genuine cross-domain chemical independence. As additional independent optimal systems are characterized, N grows and the p -value strengthens accordingly. Across these three domains, the experimentally or computationally determined operating points cluster near $\chi \approx 1$, consistent with the dynamical turnover predicted by barrier-crossing theory.

S4 Computational Implementation

The following Python implementation uses the standard library only (no external dependencies). All physical constants are NIST 2018 CODATA values. This script is the canonical computational artifact deposited at Zenodo.

Listing 1: symc_methods.py — CCE computational core

```
"""
symc_methods.py
CCE computational core (standard library only).
Author: Nate Christensen, SymC Universe Project, March 2026
"""
import math

# NIST 2018 CODATA constants
```

```

EV_TO_J    = 1.602176634e-19    # exact (2019 SI redefinition)
AMU_TO_KG  = 1.66053906660e-27
HC_EV_CM   = 1.23984193e-4      # hc in eV*cm

def chi_route_a(delta_nu_L_cm1: float, nu_cm1: float) -> float:
    """Route A: spectroscopic. delta_nu_L MUST be Lorentzian-only component
    ."""
    if delta_nu_L_cm1 <= 0 or nu_cm1 <= 0:
        raise ValueError("Frequencies must be positive.")
    return delta_nu_L_cm1 / (2.0 * nu_cm1)

def chi_route_b(tau_l_ps: float, lambda_ev: float,
                delta_q_ang: float, m_eff_amu: float) -> float:
    """Route B: solvent relaxation.
    ALWAYS call over a range of m_eff_amu. Single value is not meaningful."""
    tau_s = tau_l_ps * 1e-12
    lam_j = lambda_ev * EV_TO_J
    dq_m = delta_q_ang * 1e-10
    m_kg = m_eff_amu * AMU_TO_KG
    omega = math.sqrt(2.0 * lam_j / (m_kg * dq_m**2))
    return (1.0 / tau_s) / (2.0 * omega)

def chi_route_c(gamma_cat_ev: float, nu_cm1: float,
                gamma_int_ev: float = 0.0) -> float:
    """Route C: DFT electronic friction.
    gamma_cat from mode-projected electronic friction tensor."""
    return (gamma_cat_ev + gamma_int_ev) / (2.0 * HC_EV_CM * nu_cm1)

def chi_relative(tau_l_a_ps: float, tau_l_b_ps: float) -> float:
    """Relative chi: chi_A / chi_B = tau_L_A / tau_L_B. No m_eff needed."""
    return tau_l_a_ps / tau_l_b_ps

def test8_p_value(n_independent: int,
                  w_min: float = 0.8, w_max: float = 1.2,
                  log_min: float = -4.0, log_max: float = 1.0) -> dict:
    """Test 8 p-value. n_independent MUST be genuinely independent systems
    ."""
    log_range = log_max - log_min
    window_frac = math.log10(w_max) - math.log10(w_min)
    p_single = window_frac / log_range
    log10_p = n_independent * math.log10(p_single)
    return {"p_single_pct": round(p_single * 100, 4),
            "log10_p": round(log10_p, 3),
            "p_exact": 10 ** log10_p,
            "significant": log10_p < math.log10(0.05)}

# Sensitivity analysis
if __name__ == "__main__":
    print("=== Route B: m_eff sensitivity (TEMPO in MeCN) ===")
    for m in [50, 100, 200, 500]:
        chi = chi_route_b(tau_l_ps=0.2, lambda_ev=0.7,

```

```

        delta_q_ang=0.5, m_eff_amu=m)
    regime = ("underdamped" if chi < 0.8 else
              "near-critical" if chi <= 1.2 else "overdamped")
    print(f"   m_eff = {m:4d} amu   ->   chi = {chi:.4f}   [{regime}]")

print("\n=== Route C: Fe surface series (Maurer 2017) ===")
print(f"   Fe(111) K-promoted (gamma=0.45, nu=1415):   "
      f"chi = {chi_route_c(0.45, 1415):.3f}")
print(f"   Fe(110) (gamma=0.30, nu=2000):               "
      f"chi = {chi_route_c(0.30, 2000):.3f}")
print(f"   Fe(111) unpromoted (gamma=0.05, nu=2100):   "
      f"chi = {chi_route_c(0.05, 2100):.3f}")

print("\n=== Test 8 (N=3 independent systems) ===")
s = test8_p_value(3)
print(f"   p_single = {s['p_single_pct']:.3f}%")
print(f"   log10(p) = {s['log10_p']:.2f}")
print(f"   p_exact   = {s['p_exact']:.2e}")
print(f"   Significant (p<0.05): {s['significant']}")

```

S4.1 Reproducible calculation of the ET damping ratio

The numerical value of χ_{ET} is obtained directly from the solvent relaxation time and the barrier-crossing frequency.

$$\chi = \frac{1}{2\tau_L\Omega}$$

with

$$\tau_L = 18 \text{ ps}, \quad \Omega = 0.028 \text{ ps}^{-1}.$$

The calculation can be reproduced using the following minimal Python script:

```

import math

tau_L = 18.0
Omega = 0.028

chi = 1/(2*tau_L*Omega)

print(chi)

```

which returns

$$\chi = 0.99206349.$$

This value lies within the predicted dynamical efficiency window $[0.8, 1.2]$, confirming that the empirically observed turnover occurs at the near-critical damping condition.

S5 Extensions and Open Problems

S5.1 Multidimensional systems

When multiple coordinates couple to the reactive mode, the scalar χ is replaced by a mode-projected effective quantity:

$$\chi_{\text{eff}} = \frac{\sum_k c_k \Gamma_k}{2\sqrt{\sum_k c_k \Omega_k^2}} \quad (\text{S21})$$

This is a dominant-mode approximation that assumes one mode carries the majority of the reactive weight. The strictly rigorous expression requires a quadratic form constructed from the full friction tensor projected onto the committor subspace; Eq. (S21) is the leading-order reduction of that form when one mode dominates. The appropriate level of approximation depends on how concentrated the reactive weight is: when the dominant mode coefficient $c_{\text{max}} > 0.7$, one mode carries the reactive weight and the scalar χ_{eff} is adequate; when $0.5 < c_{\text{max}} \leq 0.7$, moderate multi-mode mixing is present and the scalar may be used with explicit qualification; when $c_{\text{max}} \leq 0.5$, strong multi-mode coupling is present and the full friction tensor projection is required. Identification of the dominant mode uses saddle-point normal-mode analysis: the mode with the largest component in the transition vector (imaginary-frequency eigenvector at the transition state).

S5.2 Machine learning integration

The CCE criterion χ can be incorporated as a descriptor in catalyst screening models. Current approaches use adsorption energy (ΔE_{ads}) and d-band center (ε_d) as primary descriptors. χ is orthogonal to both: it captures dynamical commitment rather than thermodynamic binding. A three-descriptor space (ΔE_{ads} , ε_d , χ) is expected to resolve BEP outliers that neither two-descriptor combination can separate. Implementation: add χ_{DFT} from mode-projected friction tensors to the training feature set and evaluate feature importance for outlier separation.

S5.3 Enzyme active sites

Enzyme active sites tune both Γ_{eff} through coupling to protein breathing modes and solvent channels, and Ω_{TS} through geometric strain on the substrate. The CCE prediction is that evolved enzymes exhibit $\chi_{\text{eff}} \approx 1$ for natural substrates, with off-target substrates exhibiting χ displaced from unity. Suitable test systems: cytochrome P450 (ET step, Route B measurable) and RNase A (PCET step, Route A measurable via ultrafast vibrational spectroscopy on the P–O stretch).

References

Note: References in this Supplementary Information are numbered independently of the main manuscript. All references cited here are also listed in the main manuscript reference list.

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Code (`symc_methods.py`) and all input parameters deposited at Zenodo (DOI assigned at submission). Contact: NateChristensen@SymCUniverse.com.